

MACROECONOMICS FINAL EXAM (year 4) with MARKING SCHEME

INSTRUCTIONS

- No draft paper. Everything should be written at your exam paper including calculations
- Answer all questions (weights are different and given for each problem in brackets)
- Do not change the notation
- Explain your answers clearly. There will be NO CREDIT without explanation (even if the answer is correct)
- No credits for stand-alone graphs: graphs should be well-labeled and short comments should be provided
- Do not combine the answers to the different subparts. Otherwise it is marked as the answer to the first subpart
- Time allowed: 120 minutes

1. (45 p.) Short questions

(a) 'According to the Malthusian model, since low income per worker is the result of a scarcity of land, a country with a larger land mass will have higher living standards, even in the long run.' True or false? Explain your answer.

False.

Explanation based on the graphical/algebraic analysis that compares two economies with different quantities of land given that other parameters as well as production and demographic functions are identical. (15 p.)

(b) Explain the essence of Kuznets consumption puzzle. How the modern theories of consumption resolve this puzzle?

Description of the puzzle (5 p.) Puzzle based on empirical observations that demonstrate different behavior of MPC and APC in the SR and in the LR (see Box 3.1 in UoL SG and Lectures 5-6)

Explanation based on modern theories of consumption (PIH-LCT) (10 p.)

(c) Use the model of sovereign default with limited commitment presented at the lecture to analyze the following claim: 'A country is better off if the penalty for defaulting on its foreign debts is low.'

If the claim is true then produce a proof. If the claim is false then produce a graphical counterexample and intuitive explanation.

False (see HA 12)

Intuitive explanation (5 p.)

Graphical example (10 p.)

2. (20 p.) Suppose that in the real intertemporal model with investment firms are price takers at input markets but price makers at output market.

(a) Explain the slope of output supply curve.

Explanation based on intertemporal SE (see SG section 3.11) but labour demand in this model is given by $MRPN^{REAL}$ not MPL (10 p.)

(b) Suppose that the market power is reduced due to institutional reforms. Using labour and output market diagrams analyze the impact of these reforms. Assume that wealth effects are small (but different from 0). Explain carefully/

See HA 11

Well-labeled graphs for labour and output markets with comments (10 p.)

Note: $MRPN^{REAL}$ shifts upward proportionally but it should be still below MPN (clear explanation for this shift must be provided).

3. (35 p.) Consider two economies A and B described by the Solow model. Assume that these economies have the same technologies with no technological progress and TFP equals 1. Suppose that depreciation is negligible in both economies. They also have the same initial population and the same population growth rates but economy A has two times lower saving rate than economy B. Assume both economies operate along their balanced growth paths. All markets are perfectly competitive.

(a) Using ONE graph illustrate the current position of each economy in terms of GDP per capita. At this graph show the corresponding rates of interest in every economy and explain the result.

Well-labeled Solow diagram with initial outputs and interest rates for both economies. (6 p.)

Explanation to the resulting ss (4 p.)

(b) Assume that both economies open up to foreign investment and these are the only two economies in the world. Foreign investors are able to earn a return equal to MPK per unit invested. Reproduce your graph from (a). Consider the economy that attracts foreign investment and illustrate the resulting final levels of (i) GDP per worker (denote it by y^*) and (ii) GNP per worker (denote it by y_N). Explain carefully.

Graph with GDP per worker for each economy with comments (4 p.)

GNP per worker (for each economy with comments (8 p.)

(c) Is it true that the final change in GNP per worker for the country that attracts foreign investment is ambiguous? If 'yes' then provide a detailed example. If 'no' then provide an algebraic proof.

No, algebraic proof (13 p.)